Question 2 (20 Marks)

2.1 Consider the following Java program:

```
//Program 1
public enum Color {
   RED("FF0000"), GREEN("00FF00"), BLUE("0000FF");

   private String hexCode;

   private Color(String hexCode) {
      this.hexCode = hexCode;
   }

   public String getHexCode() {
      return hexCode;
   }
   public static void main(String [] a) {
     for (Color color : Color.values())
     System.out.println((color.ordinal() + 1) + ") " + color +
                        " has code " + color.getHexCode());
   }
}

```

Based on the program above that defines an enumerated type called **`Color`**, answer the following questions:

a) What is the significance of the values "FF0000", "00FF00", and "0000FF" that are passed as arguments to the constructor of each value in the **`Color`** enumerated type?

Initialize the Hexcode field of each enum constant

(1 mark)

b) What is the purpose of the **`getHexCode()`** method in the **`Color`** enumerated type?

act as an accessor to get the return value of the Hexcode.

(1 mark)

c) What will be the output of the program?

1 RED has code FF0000
2 GREEN has code 00FF00
3 BLUE has code 0000FF

(3 marks)

d) Write a Java statement in **`main()`** method that uses the **`compareTo()`** method to display the results of a comparison of the positions of **`RED`** and **`BLUE`** in the **`Color`** enumerated type. What will the output be?

(2 marks)

System.out.println(Colour. RED . compareTo (Colour . BLUE)) ;

Output : -2

e) Consider adding the following statements in the **`main()`** method:

```
if (Color.GREEN.equals(Color.RED))
    System.out.println("GREEN and RED are two different colors.");
else
    System.out.println("GREEN and RED are the same color.");
```

The statements, however, include an error. What is the error, and how can it be fixed?

if(! Colour. GREEN. equals (Colour. RED)) **(2 marks)**

2.2 Given the directory structure as shown below:

```
C:
└── project
    ├── com
    │   ├── myexample
    │   │   ├── MyClass.java
    │   │   └── MyUtils.java
    │   └── otherexample
    │       └── OtherClass.java
    └── TestApp.java
```

a) What is the package name of **`MyClass.java`** and **`MyUtils.java`**?

com.myexample **(1 mark)**

b) What is the package name of **`OtherClass.java`**?

com. otherexample **(1 mark)**

c) In **`TestApp.java`**, how would you import all the classes in **`com.myexample`**?

import com.myexample.* ; **(1 mark)**

d) Suppose you want to move **`MyUtils.java`** to a new package called **`com.myexample.util`**. What changes would you need to make to the directory structure and the code?

(3 marks)

package com.myexample.util

2.3 Consider the Java program below and answer the following questions:

```
//Program 2
public class WrapperClass {
   public static void main(String[] args) {
      Integer num1 = 10;
      Double num2 = 3.14;

      //(a)Converting an object of a wrapper type to
      //its corresponding primitive value
      int value1 = num1.intValue();
      double value2 = num2.doubleValue();

      System.out.println("value1: " + value1);
      System.out.println("value2: " + value2);

      //(b)Adding value1 and value2 and storing the
      //result in the result variable by converting
      //it to the Double wrapper class
      Double result = Double.valueOf(value1 + value2);

      System.out.println("result: " + result);
   }
}
```

a) Fill in the blanks with an appropriate Java statement to convert an object of a wrapper type to its corresponding primitive value. **Note:** you must use a method provided in the wrapper class. What is the name of the conversion?

(3 marks)

b) Fill in the blanks with an appropriate Java statement to add **`value1`** and **`value2`** and store the result in the **`result`** variable by converting it to the **`Double`** wrapper class. **Note:** you must use a method provided in the wrapper class.

(2 marks)